

DİLBİR ÖZER

Artificial Intelligence / Deep Learning Engineer

AI/Deep Learning Engineer with hands-on experience in processing large-scale datasets, cleaning unstructured data, and optimizing data pipelines. Proficient in SQL, data modeling, and automated data validation. Skilled in ensuring data integrity, alignment with BDDK/GDPR regulations, and translating complex data landscapes into actionable insights. A detail-oriented problem solver eager to contribute to robust data governance and quality frameworks

WORK EXPERIENCE

BilSoft(CRM) / CallMetric AI AI Development

Sep 2025- Present,
Duzce, Turkey

- Developed a deep learning-based Speech Emotion Recognition (SER) system using PyTorch, Transformers, and Silero VAD.
- Managed large-scale data ingestion, cleaning, and standardization processes for speech-to-text and multi-tenant RAG pipelines to ensure high data quality and consistency
- Managed training of Speech Analysis models: Diarization, Speaker Recognition, and Emotion Extraction from audio. Worked on speech enhancement, diarization, voice activity detection (VAD), and speaker-aware audio understanding systems.
- Applied advanced hyperparameter optimization and integrated ASR outputs to maximize model performance. Utilized SQL and analytical queries to monitor and profile data flows, resolving data quality issues before production deployment
- Designed low-latency real-time speech analytics pipelines for live CRM environments.
- Multi-tenant RAG, Redis, Workers+ optional live SQL analytics, AWS, MCP, Label-studio, Open source LLM fine-tuning, model deployment, Agentic AI, Parallel Workloads, Cost optimization, EKS, S3, EC2, Load balancing, Storage optimization, Defining inbound/outbound rules, Instance Selection, document processing(OCR), Kubernetes, docker, PoC, Git Usage, BDDK/GDPR awareness, FastAPI, System Design

Norm Digital(HR) / Vinter AI Recruitment



AI Specialist

June- July 2025,
Izmir, Turkey

- Designed an AI-driven recruitment system using FAISS, and RAG (Retrieval-Augmented Generation). LangChain, Llama, Hugging Face,
- Applied Prompt Engineering and Semantic Search to enhance candidate screening accuracy.
- Implemented complex OCR-based document parsing for structured data extraction.
- Designed an AI-driven Recruitment Intelligence System for Vinter platform using LangChain, Llama, HuggingFace, FAISS, Embedding models, and RAG (Retrieval Augmented Generation) to identify the most suitable engine for candidate-job matching. Implemented OCR-based document parsing for extracting candidate information from resumes and stored structured outputs in FAISS for fast vector-based retrieval.
- Compared multiple LLM-based embeddings and RAG setups to select the most efficient architecture, ensuring optimized performance and reliable recruitment insights

Expertel SA Proceedit(FINTECH)



AI Development

Aug 2024- Feb 2025

Barcelona, Spain

- Developed scalable cloud applications using Flask and GraphQL. Created pipeline for financial time series analysis, securely scraping minute-level data using Tor and ProxyChains.
- Applied advanced models (Kolmogorov-Arnold Model, LSTM, Facebook Prophet) for high-accuracy financial forecasting.
- Engineered scalable web scraping pipelines via proxy networks, ensuring complete, accurate, and consistent financial data ingestion.
- Built a dynamic automation engine leveraging Selenium, Tor Proxy Controller, and Python multithreading to extract structured data from highly protected or JavaScript-rendered websites, optimizing efficiency and scalability in automated workflows.
- time series forecasting, anomaly detection, regression, classification

PROJECTS

- **Instagram Reach Analysis** Jun 2024 - Jul 2024, ML Project

- Conducted comprehensive data collection and preprocessing on Instagram post reach data. Performed EDA and developed/evaluated ML models to predict Instagram post reach, providing actionable insights.

- **Patient Record System** Feb 2024 - Apr 2024, DB Project

Designed and implemented a C# Windows Form application with SQL integration for Patient Registration. Developed core functionalities (record, update, delete) for efficient patient data management.

- **Smart Appointment Management System** Feb 2024 - May 2024, PM Project

- Defined scope, developed RACI matrix, and managed scheduling using Gantt Charts and WBS in EasyRedmine. Performed Software Effort Estimation (FPA, CO-COMO) and implemented Risk Monitoring & Control.

- **Film Library Management System** Dec 2023 - Jan 2024, DB/Automation

- Developed a comprehensive automation system for movie library management (add, update, search, list, rate, comment). Implemented user management features, distinguishing between standard and premium users.

TECHNICAL SKILLS

- **Deep Learning /Machine Learning /NLP:** TensorFlow, PyTorch, Keras, Hugging Face (Transformers), FAISS, Milvus (Vector DB), ASR, Diarization, Speech Emotion Recognition, LLM, Numpy, Pandas, Small Language Model, CNN , RNN, Transformer Architectures, Attention Mechanisms
- **Data Engineering/DB:** PostgreSQL, MongoDB, ETL, SQL, Web Scraping, Selenium, Tor Proxy Controller, Multithreading, MySQL, PyMuPDF
- **Programming/Web:** Python (Adv), C#, C++, HTML, CSS, Flutterflow, Flask, GraphQL, REST API
- **Tools:** Docker, Kubernetes (Basic), LangChain, RAG, Llama, GPT, Whisper, Jira, Confluence, ClickUp, EasyRedmine, Claude Code, Cursor, Notion AI, Librosa, VAD, OCR, Slack, Figma, Huggingface, Firebase, Bot Creating, Automation Engineering, DevOps, Redis, RabbitMQ, Figma, Power BI(Basic- level), Automatic1111, FishAudio, MCP
- **Model Optimization:** Quantization, PEFT (LoRA), Real-time Inference Optimization, Edge AI Concepts, Diffusion Models Concepts
- **AI Systems:** Transformer Architectures, Attention Mechanisms, CNN/RNN Fundamentals
- **Deployment:** CUDA-aware AI workflows(experienced AWS Instances selection), Low-latency AI systems, Scalable inference pipelines, Neural compression systems
- **Adaptability & Collaboration:** Rapid framework assimilation, adaptation to tech stacks, and effective cross functional teamwork.
- **Soft Skills:** Problem Solving & Analytical Thinking, Detail-oriented and quality focused, Adaptable in fastchanging project priorities, Time Management

REFERENCES

Josep Cos / Proceedit josep.cos@proceedit.com
<https://es.linkedin.com/in/josepcos>

Ferhat Sağlam / Bilsoft
+905330964800
<https://www.linkedin.com/in/ferhatsaglam/>

EDUCATION & CERTIFICATIONS

- **Bachelor of Science – Computer Engineering**, Izmir Bakircay University, Turkey (2022 – 2026, CGPA: 3.06/4.0)
- **Certifications:**
 - Python/TensorFlow for Data Science ,
 - Python Programming,
 - Automotive camp.
 - Docker:Training from basic to advanced
 - English B2
 - Defense Industry Data Marathon (22nd Place (Sayzek-Savunma Sanayi Yapay Zeka Kümelenmesi

